

Karatsuba Multiplication

The `*` operator makes multiplication easy to do in high-level programming languages such as Python and JavaScript. But low-level hardware needs a way to perform multiplication using more primitive operations. We could multiply two integers using only addition with a loop, such as in the following Python code to multiply $5678 * 1234$:

```
>>> x = 5678
>>> y = 1234
>>> product = 0
>>> for i in range(x):
...     product += y
...
>>> product
7006652
```

However, this code doesn't scale efficiently for large integers. *Karatsuba multiplication* is a fast, recursive algorithm discovered in 1960 by Anatoly Karatsuba that can multiply integers using addition, subtraction, and a precomputed multiplication table of all products from single-digit numbers. This multiplication table, shown in [Figure 5-5](#), is called a *lookup table*.

Our algorithm won't need to multiply single-digit numbers because it can just look them up in the table. By using memory to store precomputed values, we increase memory usage to decrease CPU runtime.